

Digital Evidence Collection and Investigation

Tool: FTK Imager

Theory

FTK Imager is a free digital forensics tool used to create exact bit-by-bit forensic images of storage devices without altering the original data. It also allows investigators to browse file systems, recover deleted files, and examine browser artifacts such as history and cookies. A forensic image preserves evidence integrity, and MD5/SHA1 hash verification ensures the image is identical to the original source.

Steps

1. Download FTK Imager from <https://www.exterro.com/ftk-imager> and install it on a Windows machine. Run the installer as Administrator and follow the setup wizard.
 2. Launch FTK Imager. Go to File menu and select Add Evidence Item.
 3. Select Physical Drive as the source type and click Next. Choose the target drive or USB from the list and click Finish. The drive contents will appear in the Evidence Tree panel on the left.
 4. Browse the file system in the Evidence Tree. Expand the drive to navigate through folders. Click any file to preview its contents in the right panel.
 5. To find browser artifacts, navigate to: Users > [Username] > AppData > Local > Google > Chrome > User Data > Default. Look for files named History, Cookies, and Login Data.
 6. Right-click the History file and select Export Files. Save it to a folder on the Desktop for analysis.
 7. To create a forensic image of the drive, go to File > Create Disk Image. Select Physical Drive as source, choose the drive, then click Add to set the destination. Select E01 format, enter case details, choose output folder, enable Verify images after they are created, and click Start.
 8. After imaging is complete, FTK Imager displays the MD5 and SHA1 hash values for both the source and the image. Confirm both hashes match — this verifies the image is an exact copy and evidence integrity is maintained.
 9. To capture RAM, go to File > Capture Memory. Choose a destination folder and click Capture Memory. This saves a .mem file containing the contents of physical memory at the time of capture.
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Conclusion

FTK Imager successfully created a forensic image of the storage device and extracted browser artifacts including history and cookies. Hash verification confirmed the image integrity, making the collected evidence suitable for use in a formal investigation.